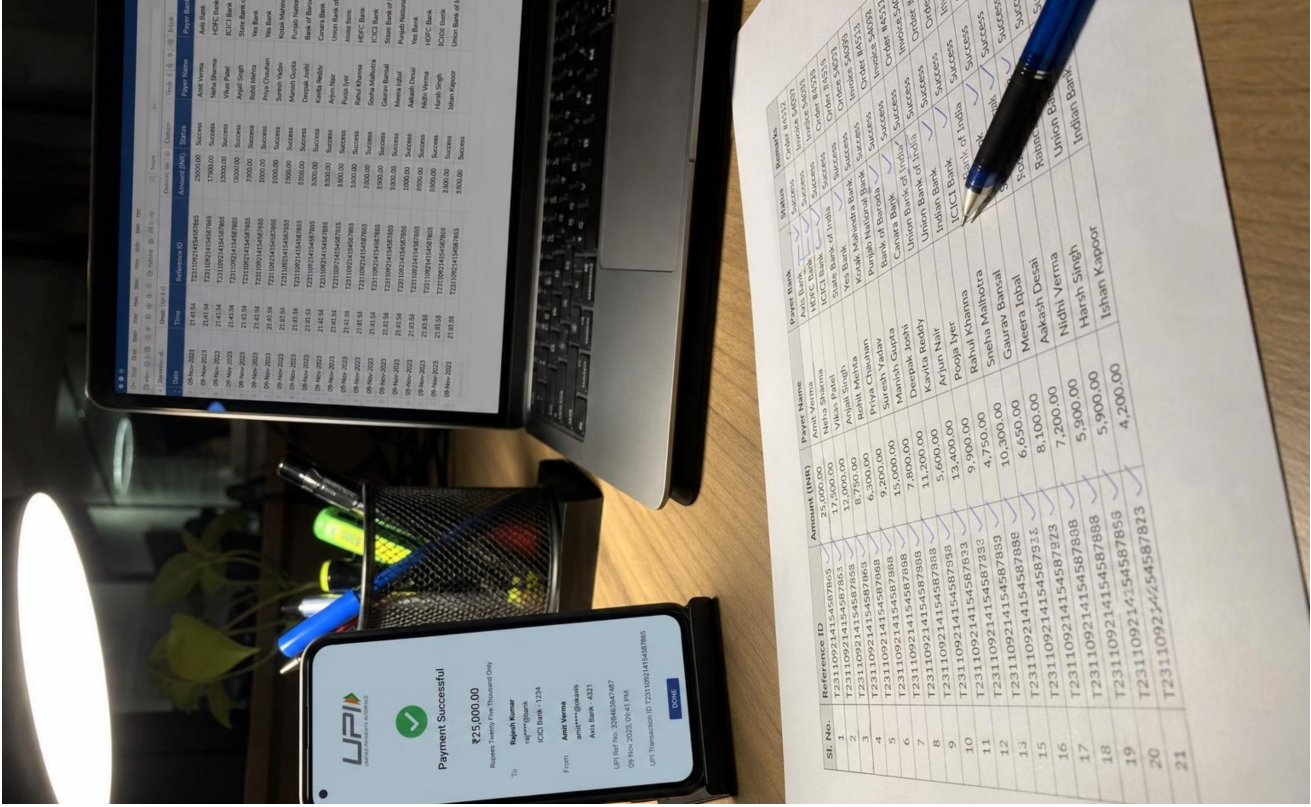


FinTech UPI Fraud Intelligence



Track 1: AgentIQ Datathon Submission

Presenter: Zaaki Mohammed

Team:

EC Aiswarya

Bhavya S Nair

Zaaki Mohammed

Moving beyond isolated fraud to detect organized criminal networks.

Engineering the Foundation

20,000

Raw UPI transactions

5-way join · 100% preserved

01

Processed 20,000 raw UPI transactions.

03

Executed a flawless 5-way join across all datasets (KYC, Merchants, Chargebacks).

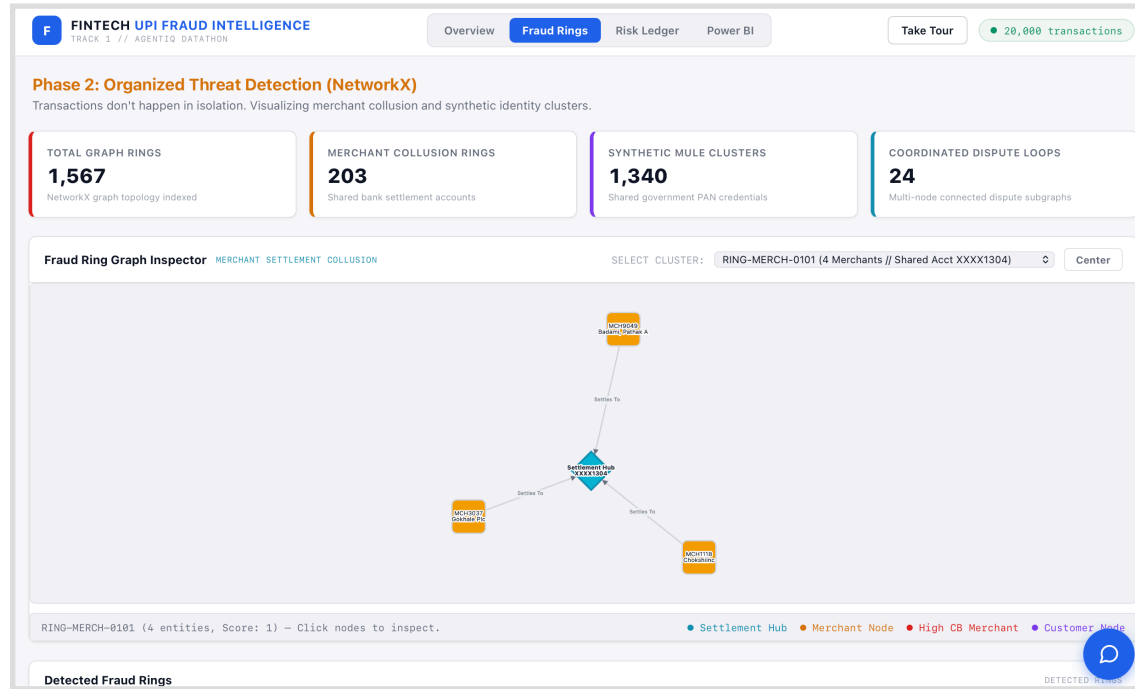
02

Utilized Pandas and Regex to clean and normalize mismatched user and merchant IDs.

04

Achieved 100% data preservation without recklessly dropping missing rows.

Uncovering the Hidden Networks



Standard SQL queries miss organized fraud. We used NetworkX.

Modeled the ecosystem as a bipartite graph (Users -> Merchants).

Discovered 203 Merchant Collusion Rings sharing identical bank settlement accounts.

Uncovered 1,340 Synthetic Identity Clusters operating under the exact same PAN credentials.

203

Merchant Collusion Rings

Identical bank settlement accounts

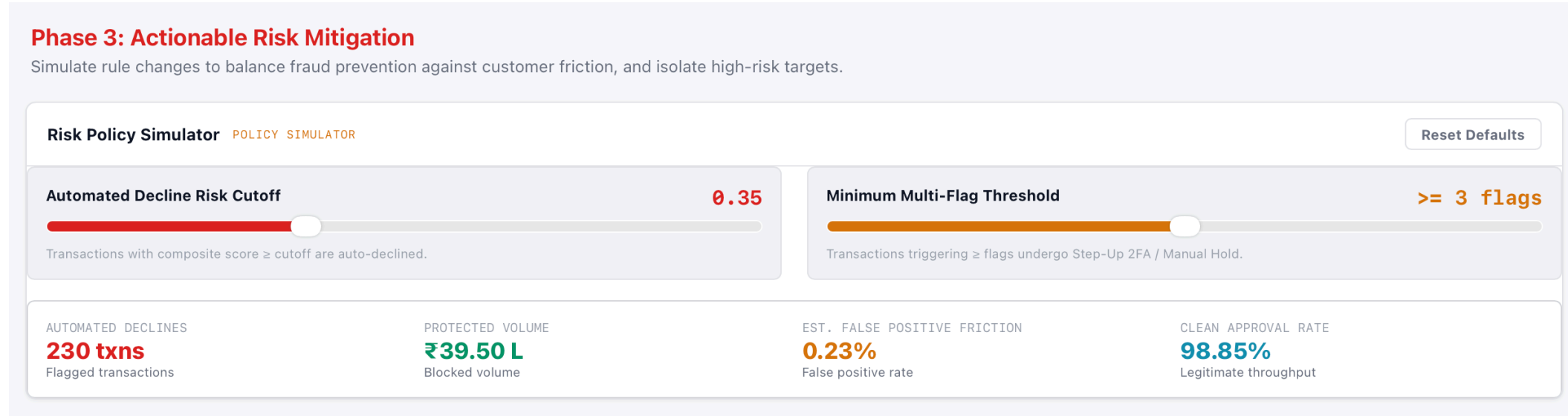
1,340

Synthetic Identity Clusters

The exact same PAN credentials

The Risk Policy Simulator

Moving beyond static charts to actionable risk mitigation.



Risk Policy Simulator

Built in custom HTML/JS.

Executives instantly balance fraud prevention against false-positive customer friction.

Power BI Dashboard

Enterprise analyst reporting.

A fully integrated Power BI dashboard drives deep-dive analyst reporting.

Securing the 30-Point AI Bonus

01	02	03	04
Model layer	Query layer	Visual layer	Trust layer
Powered by Groq (Qwen 27B) and LangChain.	Translates natural English into secure DuckDB SQL queries in real-time.	Dynamically renders Matplotlib charts (Bar, Line, Pie, Scatter) based on intent.	Every answer ships with a transparent SQL trace UI, so each query is fully auditable end to end.

Layer	Component	Guarantee
Inference	Groq · Qwen 27B	Real-time
Orchestration	LangChain	Intent routing
Execution	DuckDB SQL	Secure
Audit	SQL trace UI	Fully auditable

